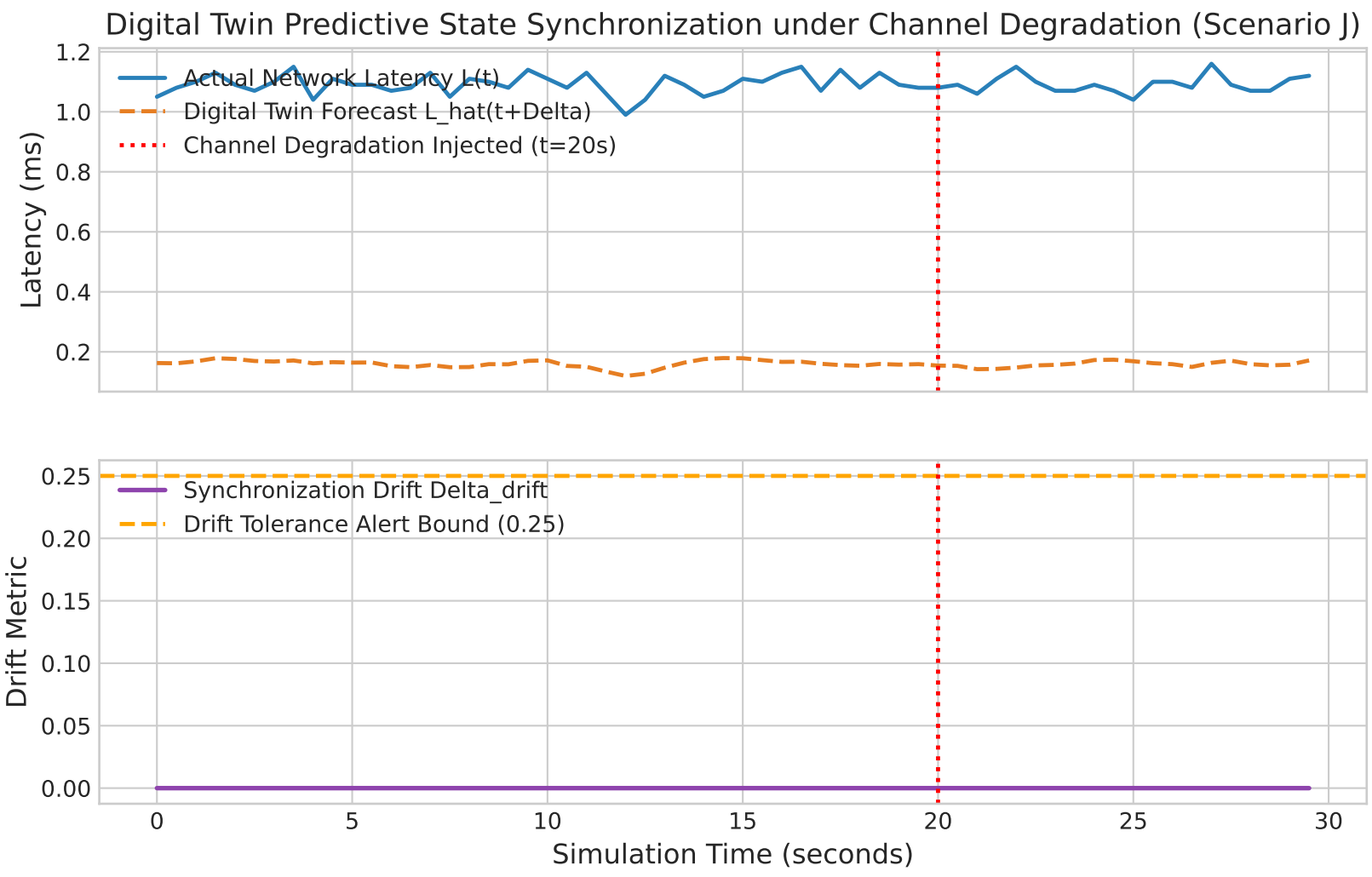
Digital Twin Predictive State Synchronization under Channel Degradation (Scenario J)
Actual Network Latency L(t)
Digital Twin Forecast L_hat(t+Delta)
Channel Degradation Injected (t=20s)
Latency (ms)
1.2
1.0
0.8
0.6
0.4
0.2
Synchronization Drift Delta_drift
Drift Tolerance Alert Bound (0.25)
Drift Metric
0.25
0.20
0.15
0.10
0.05
0.00
0
5
10
15
20
25
30
Simulation Time (seconds)